



Efficacy and safety of a novel monopolar radiofrequency device for skin tightening: a multicenter, randomized, 6-month, assessor blind, positive parallel-controlled, non-inferiority trial

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Abstract

Skin laxity and static wrinkles cause aesthetic concerns. Monopolar radiofrequency (RF) could produce measurable improvement. This study aims to evaluate the efficacy and safety of a novel monopolar RF device, YOUMAGIC, for skin tightening and rejuvenation treatment. In this multicenter, 6-month, assessor blind, positive parallel-controlled study, subjects were randomized to receive a single treatment with YOUMAGIC or Thermage (CPT; Solta Medical). Efficacy was mainly determined by the effective rate of facial wrinkle improvement [Global Aesthetic Improvement Scale (GAIS)] 90 days after treatment. Safety assessments were conducted through structured inquiries, adverse event diaries maintained by subjects, laboratory tests, and monitoring of vital signs. Of the 230 enrolled subjects, 212 were included in the final analysis. The proportions of subjects who achieved ≤ 3 -grade on the GAIS were 100% in the treatment group and 98.10% in the control group 90 days after treatment. No significant differences were found between the treatment and control groups. No subjects experienced any serious adverse events. In conclusion, the novel monopolar RF device YOUMAGIC is clinically safe and effective for skin tightening and is not inferior to the regular RF therapeutic device.

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Keywords Monopolar radiofrequency · Skin tightening · YOUMAGIC · Thermage

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Introduction

Skin laxity and static wrinkles are prevalent aesthetic concerns that arise with age and exposure to ultraviolet light. The gradual reduction in human dermal collagen content leads to skin flabbiness, fine wrinkles, pigmentation issues, changes in skin texture, and capillary dilation. These issues variably impact individuals' satisfaction and confidence in their appearance.

Numerous invasive and non-invasive procedures have been developed and are currently available to tackle these concerns. Invasive methods like rhytidectomy carry the potential risks associated with anesthesia and prolonged recovery periods. As a result, non-invasive skin rejuvenation procedures have gained popularity among patients seeking practical solutions for skin lifting and wrinkle reduction.

Over the past decade, numerous energy-based devices, such as Radiofrequency (RF) technology, have gained recognition for their effective skin-tightening capabilities. The

tissue tightening mechanism of these procedures is similar, involving the transmission of heat energy into the dermal layer. This process results in partial collagen denaturation, stimulating the production of new collagen and elastin while safeguarding the integrity of the epidermis [1].

RF is an electromagnetic wave ranging from 3 kHz to 300 GHz. When RF electromagnetic waves pass through tissues, they generate heat effects due to their impedance, heating of the dermis while keeping the epidermal temperature below 45 °C. This leads to collagen contraction and denaturation, triggering a wound-healing response and promoting the remodeling and regeneration of collagen. Monopolar RF comprises a main electrode head and an electrode plate attached to the skin. The current flows from the electrode head through the human body to the electrode plate. Due to the high temperature near the electrode head, the epidermis needs to be cooled to maintain its normal temperature while the dermis is heated. Skin laxity improvement is achieved through immediate and long-term thermal effects [2]. Monopolar RF could produce measurable improvement in skin laxity with an acceptable complication profile and is currently considered the gold standard for the non-invasive treatment of skin laxity [3, 4].

Recently, a novel monopolar RF treatment device, YOUMAGIC, was developed in China for the treatment of skin laxity. The radiofrequency generator also generates a 6.78 MHz radiofrequency field. The refrigeration level is adjustable from 1 to 4, which could be adjusted according to the patient's skin temperature during the treatment. Single radiofrequency time was 0.7–1.5 s with nine adjustable levels; the short pulses can quickly accumulate heat and reduce the treatment time, and long pulses could provide a slow and gentle heating process and thus relieve pain. The resistance value of the last treatment site could be measured and displayed, which helps doctors evaluate the skin status of patients and provides a reference for selecting treatment parameters.

Therefore, in this randomized, assessor blind, multicenter, positive-controlled clinical trial, we aimed to evaluate the effects and safety of YOUMAGIC in treating skin laxity.

Materials and methods

Overview

The study aimed to assess the effectiveness and safety of a novel monopolar RF device, YOUMAGIC, for skin tightening treatment. The 212 subjects from four hospitals, consisted of 12 males and 200 females aged 30 to 60 years with Fitzpatrick skin types II–V and mild to moderate

skin laxity, were included in this study. Subjects were randomly assigned in a 1:1 ratio to the novel RF device (YOUMAGIC, 107 patients) or control (Thermage CPT System, 105 patients), receiving a single treatment. Before their participation, all individuals received detailed information about the study protocol and provided signed informed consent. Digital photographs and VISIA images were captured at baseline, 30, 90, and 180-day post-treatment to evaluate the efficacy of the device for skin tightening. Pain levels were assessed immediately after treatment using a pain measurement scale. Any adverse events (AEs) reported by participants during the study period or noted by the research team were meticulously documented and reviewed by a dermatologist.

Study plan

This clinical trial took place from May 2023 to February 2024. The study was conducted over a period of 6 months, involving four visits. The first visit, considered the baseline visit, encompassed the screening procedure, obtaining informed consent, and administering the monopolar RF treatment. Subsequent visits, occurring at 30, 90, and 180 days after the baseline visit, were designated as follow-up visits during which the research team evaluated the subjects to assess the outcomes post-treatment. All images were captured using digital photography and the VISIA system (Canfield Scientific Inc., Parsippany, NJ). Global Aesthetic Improvement Scale (GAIS) and Fitzpatrick Wrinkle Classification System (FWCS) were used to evaluate the efficacy. The pain was assessed by numerical rating scale (NRS) immediately after treatment. Any procedure-related side effects were documented and examined.

Inclusion and exclusion criteria

Inclusion criteria were healthy males and females aged 30–60, diagnosed with mild to moderate sagging skin on the face area (FWCS 3–6 point), defined as Fitzpatrick skin types II–V, and opted for a skin tightening procedure. Subjects were excluded if they had a history of severe skin allergies and currently experiencing skin allergy flare-ups; Sun exposure within the past week; botulinum toxin, fat injections, or other anti-wrinkle treatments in the treatment area within the past 6 months; monopolar radiofrequency or ultrasonic scalpel treatment or having received injections of hyaluronic acid, polylactic acid, or PCL facial fillers in the treatment area within the past 1 year; facial lifting surgery or any facial plastic surgery within the past 1 year; bipolar or multipolar radiofrequency treatments, ablative laser treatments, or other photonic treatments such as facial photo rejuvenation within the past 6 months; deep chemical peel

within the past 1 month; inflammatory reactions, significant scars, hemangiomas, severe acne, or other skin lesions that may affect the interpretation of study results in the treatment area; systemic or localized infections; implantable pacemakers, implantable cardioverter-defibrillators, or any other type of implantable electronic devices; metallic implants in the treatment area, such as dental implants; severe obesity ($\text{BMI} > 30 \text{ kg/m}^2$); History of facial nerve paralysis. Women who are pregnant, lactating, or planning to become pregnant during the study period were also excluded.

Treatment

The treated area was gently cleansed with a mild facial cleanser and a topical anesthetic cream (5% lidocaine) was applied for 30 min. Before the initiation of treatment, the subject was asked to adopt a supine position, the return electrode was positioned beneath each patient's back to set a monopolar circuit, and RF gel was applied to the treated area to serve as a coupling medium. A 6.78-MHz RF device (YOUMAGIC or Thermage CPT, 900 shots, energy level 2.5–5.0) was used for treatment. During treatment, the treatment head fits on the surface of the body and uses a cooling agent on the body surface. When RF waves are applied, a reverse temperature gradient is created in the skin, maintaining the epidermal temperature below 45°C and heating the dermal tissue. The procedure lasted 40–60 min, and the RF gel was wiped off and cleaned after treatment.

Efficacy evaluation

The primary efficacy endpoint of this study was to evaluate the effective rate of facial wrinkle improvement at 90 days after treatment. The effective rate was characterized by the proportions of subjects who achieved ≤ 3 -grade on the GAIS by three independent blinded evaluators using the 5-point GAIS (1=Very much improved; 2=Much improved; 3=Improved; 4=No change; 5=Worse), mainly focused on the nasolabial fold.

Secondary efficacy endpoints were comprised of assessments of facial wrinkle improvement rate on GAIS at 30 and 180 days after treatment, FWCS at 30, 90, and 180 days after treatment by independent blinded evaluators, as well as GAIS by both subjects and evaluators at 30, 90, and 180 days after treatment, and NRS at the treatment day. NRS was classified as no pain (0), mild pain (1–3), moderate pain (4–6), and severe pain (7–10).

Safety evaluation

Safety was assessed by inquiries, subject diary entries of adverse events, laboratory tests, and vital sign checks at each

follow-up. Laboratory tests, including complete blood counts, general chemical tests, blood coagulation tests, and blood/urine pregnancy test, were undertaken at the screening visit and then at 180 days (the first two tests). Adverse events (AEs) were evaluated through self-reports and physical examinations at every follow-up, as well as subject diary entries recording adverse reactions within 7 days after treatment [5].

Statistical analysis

The primary endpoint, the proportion of subjects who achieve ≤ 3 -grade on the GAIS, is expected to be 94% in the control group based on previous studies [6–9]. It is assumed that the therapeutic effect of the treatment group is equivalent to that of the control group, and the non-inferiority margin value is -10%. Thus, calculations showed that at least 89 subjects per group were required to achieve a power of 80% at a 2.5% significance level for assessment of this endpoint. Based on the assumed dropout rate of 15%, a total of 212 subjects were included.

For the primary endpoint assessment, a two-sided 95% confidence interval was calculated for the difference in the improvement rates. Non-inferiority was concluded if the lower limit of the confidence interval was above the non-inferiority margin value (-10%). The t-tests or Wilcoxon signed-rank tests were used to analyze continuous variables. Categorical variables were compared using χ^2 tests or Fisher's exact tests. Data were analyzed using SAS version 9.4.

Results

Demographics and baseline characteristics of the subjects

Of the 230 enrolled subjects, 18 were excluded because they did not meet the inclusion criteria, and 212 were included in the full analysis set (FAS) (Fig. 1). In FAS, the mean age was 43.13 years, 94.34% were female, and the mean BMI was 22.28 kg/m^2 . The average age and BMI of the subjects had no statistical differences between the two groups (Table 1).

Efficacy

Primary efficacy endpoint

The proportions of subjects who achieved ≤ 3 -grade on the GAIS of facial wrinkles 90 days after treatment in the treatment group and the control group were 100% and 98.10% in FAS ($p=0.24$) and 100% and 98.10% in the per-protocol set (PPS). No significant differences were found between the treatment and control groups (Table 2). FAS and PPS

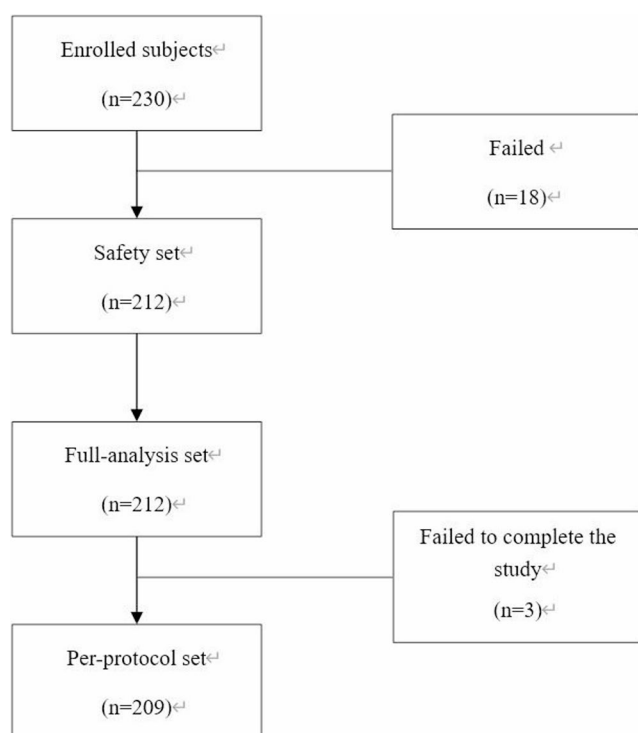


Fig. 1 Flowcharts of the subjects included in this study

Table 1 Demographics and baseline characteristics of the subjects

Variable	Total (n=212)	YOU- MAGIC (n=107)	Thermage (n=105)	P value
Age, years	43.13±7.91	43.51±7.77	42.73±8.06	0.47
Female, n (%)	200(94.34)	103(96.26)	97(92.38)	0.22
BMI, kg/m ²	22.28±2.55	22.29±2.54	22.27±2.57	0.95

BMI, body mass index, shown as mean (SD)

reached the same conclusion for the primary efficacy endpoint. A non-inferiority test was conducted on the effective rate of facial wrinkle improvement at 90 days post-treatment in FAS and PPS. The results showed that the difference between the two groups was -1.90% and -1.94% , and the two-sided 95% confidence interval of the difference was $(-1.83, 6.68\%)$ and $(-1.83, 6.81\%)$. The two-sided 95% confidence interval did not exceed the pre-defined non-inferiority margin of -10% , and the non-inferiority hypothesis was established (Table 3). Logistic regression was used to conduct a center effect analysis on the effective rate of facial wrinkles at 90 days post-treatment in FAS and PPS. This clinical study showed no significant center effect ($P=0.40$) or group effect ($P=0.19$).

Although no difference was found between the two groups for the effective rate, it is worth mentioning that the proportion of 1- and 2-grade on GAIS of facial wrinkles was 73.8% in the YOUMAGIC group, which was significantly higher than the 59.0% in the control group. (Fig. 2)

Table 2 Effective rate at 90 days post-treatment

Analytic set	Total	YOUMAGIC	Thermage	P value
FAS (%)	210/212 (99.06)	107/107 (100.00)	103/105 (98.10)	0.24
PPS (%)	207/209 (99.04)	106/106 (100.00)	101/103 (98.06)	0.24

The effective rate was characterized by the proportions of subjects who achieved ≤ 3 -grade on the GAIS of facial wrinkles

Table 3 Non-inferiority test of the effective rate at 90 days post-treatment

Analytic set	Effective rate at 90 days post-treatment		Two-sided 95% CI	<i>P</i> value §	<i>P</i> value ¶
	YOUMAGIC	Thermage			
FAS (%)	100.00	98.10	1.90(-1.83,6.68)	0.40	0.19
PPS (%)	100.00	98.06	1.94(-1.83,6.81)	0.41	0.19

CI, confidence interval. § central effect, ¶ group effect

Secondary efficacy endpoint

The effective rates of facial wrinkle improvement 30 and 180 days post-treatment were 90.65% and 96.23% in the treatment group and 90.29% and 91.18% in the control group, respectively. No significant differences were found between the treatment and control groups. FWCS at 30, 90, and 180 days after treatment was 4.39 ± 1.04 , 3.94 ± 1.07 , and 4.01 ± 0.99 in the treatment group, and 4.22 ± 1.04 , 3.81 ± 1.13 , and 3.88 ± 1.04 in the control group, respectively. No significant differences were found between the treatment and control groups. In addition, GAIS scores by subjects and evaluators were assessed at 30, 90, and 180 days post-treatment, and no significant differences were found (Table 4). NRS on the treatment day was evaluated, and no significant differences were found between the two groups (Table 5).

Safety

In this clinical trial, the safety set (SS) consisted of 212 cases. A total of 103 cases resulted in 256 adverse events, of which 109 adverse events occurred in 52 subjects related to the treatment. Most of the AEs were pain, erythema, thermal burn, edema, etc. Treatment area AEs were reported by 25/107 (23.36%) in the treatment group and 27/105 (25.71%) in the control group. Pain was the most common side effect (20/212, 9.43%), followed by erythema (17/212, 8.02%), thermal burn (16/212, 7.55%), and edema (13/212, 6.13%) (Table 6).

The majority of AEs were mild or moderate in severity. All adverse reactions resolved spontaneously without special treatment. Notably, erythema reported by participants

Fig. 2 Clinical photographs before and after treatment in the treatment group (a, b, c) and control group (d, e, f)



representing a similar shape to the treatment head area or lasting more than 24 h after RF device treatment was defined as “thermal burn”, and most of the “thermal burns” were determined as “mild” by researchers. Two subjects experienced serious adverse events (pregnancy and open injury of the right lower limb), both from the control group, which were determined to be unrelated to the clinical trial.

Discussion

This study demonstrates improvements in skin tightening and rejuvenation after single treatments with a novel RF device, YOUMAGIC, compared with the Thermage CPT device. Remarkable improvements from baseline were observed in both groups at 30, 90, and 180 days post-treatment, with no significant differences noted.

RF has shown significant efficacy in facial lifting, improving skin texture, and reducing wrinkles. As a non-invasive technique, RF is known for its precise therapeutic effects, safety, minimal adverse reactions, and short downtime. The earliest technology widely used in clinical practice is monopolar radiofrequency, which was approved by the FDA in 2002 for treating periorbital wrinkles, in 2004 for facial wrinkles, and in 2005 for all types of wrinkles.

In the field of skin anti-aging, monopolar radiofrequency is currently considered the gold standard for the non-invasive treatment of skin laxity. In the early 21st century, dermatologists evaluated the efficacy of periorbital wrinkles using the earliest device (ThermaCool TC) and found significant improvements, both objective and subjective, in over 80% of patients 6 months after a single treatment [8, 10]. Except for the periorbital area, RF treatment improved the lower face and neck. Nahm et al. [11] found

Table 4 Secondary efficacy endpoints

Efficacy evaluation	YOUNG				Thermage				P value			
	30 days	90 days	180 days	30 days	90 days	180 days	30 days	90 days	180 days	30 days	90 days	180 days
effective rate (%)	97/107(90.65)	/	102/106(96.23)	93/103(90.29)	/	93/102(91.18)	0.93	/	93/102(91.18)	0.93	/	0.13
FWCS	4.39±1.04	3.94±1.07	4.01±0.99	4.22±1.04	3.81±1.13	3.88±1.04	0.24	3.81±1.13	3.88±1.04	0.24	0.37	0.37
GAIS by subjects	18/107(16.82)	25/107(23.36)	26/107(24.53)	10/105(9.71)	20/105(19.05)	14/105(13.73)	0.37	20/105(19.05)	14/105(13.73)	0.37	0.29	0.26
Much improved (%)	31/107(28.97)	32/107(29.91)	38/107(35.85)	38/105(36.89)	44/105(41.90)	40/105(39.22)						
Improved (%)	53/107(49.53)	49/107(45.79)	36/107(33.96)	49/105(47.57)	40/105(38.10)	42/105(41.18)						
No change (%)	5/107(4.67)	1/107(0.93)	6/107(5.66)	6/105(5.83)	1/105(0.95)	6/105(5.88)						
Worse (%)	0/107(0)	0/107(0)	0/107(0)	0/105(0)	0/105(0)	0/105(0)						
GAIS by evaluators	13/107(12.15)	14/107(13.08)	5/107(4.72)	13/105(12.62)	16/105(15.24)	10/105(9.80)	0.80	16/105(15.24)	10/105(9.80)	0.80	0.72	0.16
Much improved (%)	55/107(51.40)	52/107(48.60)	45/107(42.45)	48/105(46.60)	46/105(43.81)	45/105(44.12)						
Improved (%)	37/107(34.58)	40/107(37.38)	48/107(45.28)	38/105(36.89)	40/105(38.10)	45/105(44.12)						
No change (%)	2/107(1.87)	1/107(0.93)	8/107(7.55)	4/105(3.88)	3/105(2.86)	2/105(1.96)						
Worse (%)	0/107(0)	0/107(0)	0/107(0)	0/105(0)	0/105(0)	0/105(0)						

The effective rate was characterized by the proportions of subjects who achieved ≤ 3 grade on the GAIS of facial wrinkles; FWCS, Fitzpatrick Wrinkle Classification System; GAIS, Global Aesthetic Improvement Scale

Table 5 NRS assessment

NRS	YOUMAGIC (<i>n</i> = 107)	Thermage (<i>n</i> = 105)	<i>P</i> value
no pain (%)	5 (4.67)	7 (6.67)	0.58
mild pain (%)	41 (38.32)	48 (45.71)	
moderate pain (%)	53 (49.53)	43 (40.95)	
severe pain (%)	8 (7.48)	7 (6.67)	

NRS, numerical rating scale, no pain (0), mild pain (1–3), moderate pain (4–6), and severe pain

Table 6 Summary of main adverse events

Variable	Total (<i>n</i> = 212)	YOU- MAGIC (<i>n</i> = 107)	Thermage (<i>n</i> = 105)	<i>P</i> value
Adverse events (%)	52 (24.53)	25(23.36)	27(25.71)	0.69
Pain (%)	20 (9.43)	11 (10.28)	9 (8.57)	
Erythema (%)	17(8.02)	7(6.54)	10 (9.52)	
Thermal burn (%)	16 (7.55)	8 (7.48)	8 (7.62)	
Edema (%)	13 (6.13)	6 (5.61)	7 (6.67)	
Pruritus (%)	8 (3.77)	3(2.80)	5 (4.76)	
Pigmentary abnormal- ity (%)	6 (2.83)	1 (0.93)	5 (4.76)	
Hypoesthesia (%)	5 (2.36)	3 (2.80)	2 (1.90)	
Sensation variation (%)	5 (2.36)	1 (0.93)	4 (3.81)	

a mean decrease of 22.6% in jowl surface area on the lower part of the face 3 months after treatments. Bogle et al. [12] found that 92% of patients had a measurable improvement in overall appearance after treatment of the lower face and neck at the 6-month follow-up visit. In another study conducted in the Asian group with the same device, significant improvements were observed in nasolabial folds, marionette lines, and jawline ptosis at 3 months post-treatment, with the best results at 6 months [9]. Abraham et al. [13] conducted a prospective study in 35 healthy adults with moderate facial aging and found all skin parameters (laxity, wrinkles, clarity, pore size) were improved 12 weeks post-treatment. Subsequently, a retrospective study [6] was conducted to evaluate the skin-tightening effects of a new-generation device (Thermage CPT). After two treatment sessions at 10 months, 80% of patients noted mild to moderate improvement in laxity and skin texture, with erythema and edema resolving in 1 to 7 days.

At this point, our study investigated the potential of a novel monopolar RF device to stimulate collagen remodeling, thereby promoting skin rejuvenation with reduced complications. We found that the effectiveness of the novel RF device in skin tightening and rejuvenation was not inferior to that of the Thermage CPT system, and the effect was maintained for up to 6 months. Furthermore, this research assessed the safety aspects of the new RF machine. No severe adverse events related to the treatment were observed during treatment, and the pain scores were similar in the two groups. Regarding limitations, the efficacy evaluation of our

study relies primarily on subjective outcome measures and lacked objective instrumental or histological assessments. In addition, our study population was predominantly female (approximately 94%), with only 12 male participants, which might limit the generalizability of the results.

In conclusion, the novel monopolar RF device YOUMAGIC has been demonstrated to be clinically safe and effective for skin tightening. This innovative technology is a viable option for patients seeking non-invasive skin tightening and rejuvenation, offering promising outcomes.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s10103-026-04841-4>.

Author contributions All authors contributed to the study conception and design. Material preparation, data collection, and analysis were performed by Ting Li, Qingbiao Wa, Zhenfeng Liu, Bin Yang, and Yan Yan. The first draft of the manuscript was written by Zhongshuai Wang, and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript. Corresponding author: Yan Yanyanyan@psh.pumc.edu.cn.

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Data availability The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval All patients provided written informed consent and permission to publish. All procedures were approved by Ethics Committee. Our study was performed in accordance with the Helsinki Declaration of 1964, and its later amendments.

Competing interests The authors declare no competing interests.

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